Comprehensive Supplemental Material for "Observer-Dependent Temporal Asymmetry: From Quantum Erasure to Gravitational Time"

Sheng-Kai Huang[1,*]

[1] *Independent Researcher*

(Dated: March 12, 2026)

CONTENTS

I. S8. Proof of Theorem 1 (Monotonicity) 3

A. Statement 3

B. Full proof via DPI on nested channels 3

C. Gap between optimal recovery and Petz recovery 4

D. Worked example: qubit dephasing with partial environment 5

II. S9. Proof of Theorem 2 (Operational Threshold) 6

A. Statement 6

B. Fuchs–van de Graaf inequality 6

C. Simplification for small τ 6

D. Chernoff–Stein lemma application 7

III. S10. Proof of Theorem 3 (Subadditivity) 8

A. Statement 8

B. Proof from Theorem 1 8

C. Information-theoretic interpretation 8

IV. S11. Evidence for Conjecture 1 (Complementary Uncertainty) 9

A. Statement of the conjecture 9

B. Qubit MUB calculation 9

C. Verification of the bound 10

D. Why the general proof is hard 11

V. S12. Quantum Eraser Worked Example 12

A. Setup 12

B. Observer 1: Signal-only 12

C. Observer 2: Signal + Idler (total observer) 14

D. Observer 3: Idler-only 14

E. Summary 15

VI. S13. Black Hole Complementarity Calculation 15

* akai@fawstudio.com

A. Infalling observer 15
B. External observer 15
C. Consistency (no contradiction) 16

VII. S14. Observer Hierarchy Table 17
A. τ values and justification 17
B. Justification for each estimate 17
C. The hierarchy is monotone 18

References 18

I. S8. PROOF OF THEOREM 1 (MONOTONICITY)

We provide the full proof of monotonicity of τ under observer refinement, carefully distinguishing the optimal-recovery and Petz-recovery cases.

A. Statement

Theorem 1 (Monotonicity of τ under observer refinement)**.** *If $O_1 \leq O_2$ (i.e., $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_2}$), then*

$$\tau_{O_1}^{\mathrm{opt}} \geq \tau_{O_2}^{\mathrm{opt}}\,, \tag{1}$$

where $\tau_O^{\mathrm{opt}} = 1 - \max_{\mathcal{R}} F(\rho, \mathcal{R} \circ \mathcal{N}_O(\rho))$ is the temporal asymmetry under optimal recovery.

B. Full proof via DPI on nested channels

Proof. Since $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_2}$, there exists a CPTP map (conditional expectation) $\mathcal{E}_{12} : \mathcal{A}_{O_2} \to \mathcal{A}_{O_1}$ such that $\mathcal{N}_{O_1} = \mathcal{E}_{12} \circ \mathcal{N}_{O_2}$. This factorization is a consequence of the lattice structure of von Neumann subalgebras: when $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_2}$, the restriction from the larger algebra to the smaller one is a normal conditional expectation [1], which is automatically CPTP.

Step 1 (DPI for relative entropy).—By the data processing inequality (DPI) applied to

$\mathcal{E}_{12}$:

$$\begin{aligned} &D(\mathcal{N}_{O_2}(\rho)\|\mathcal{N}_{O_2}(\sigma)) \\ &\quad \geq D(\mathcal{E}_{12}(\mathcal{N}_{O_2}(\rho))\|\mathcal{E}_{12}(\mathcal{N}_{O_2}(\sigma))) \\ &\quad = D(\mathcal{N}_{O_1}(\rho)\|\mathcal{N}_{O_1}(\sigma))\,. \end{aligned} \tag{2}$$

Therefore, the entropy production satisfies

$$\begin{aligned} \Sigma_{O_1} &= D(\rho\|\sigma) - D(\mathcal{N}_{O_1}(\rho)\|\mathcal{N}_{O_1}(\sigma)) \\ &\geq D(\rho\|\sigma) - D(\mathcal{N}_{O_2}(\rho)\|\mathcal{N}_{O_2}(\sigma)) = \Sigma_{O_2}\,. \end{aligned} \tag{3}$$

Step 2 (Optimal fidelity monotonicity).—Define the optimal recovery fidelity:

$$F_{\text{opt}}(\mathcal{N}) \equiv \max_{\mathcal{R}:\,\text{CPTP}} F(\rho,\, \mathcal{R}\circ\mathcal{N}(\rho))\,. \tag{4}$$

For any recovery map $\mathcal{R}_1 : \mathcal{A}_{O_1} \to \mathcal{A}_{\text{total}}$, define $\mathcal{R}_1' = \mathcal{R}_1 \circ \mathcal{E}_{12} : \mathcal{A}_{O_2} \to \mathcal{A}_{\text{total}}$. Then $\mathcal{R}_1'$ is a valid CPTP recovery map from $\mathcal{A}_{O_2}$, and

$$F(\rho,\, \mathcal{R}_1\circ\mathcal{N}_{O_1}(\rho)) = F(\rho,\, \mathcal{R}_1'\circ\mathcal{N}_{O_2}(\rho))\,. \tag{5}$$

Taking the maximum over $\mathcal{R}_1$:

$$\begin{aligned} F_{\text{opt}}(\mathcal{N}_{O_1}) &= \max_{\mathcal{R}_1} F(\rho,\, \mathcal{R}_1\circ\mathcal{N}_{O_1}(\rho)) \\ &= \max_{\mathcal{R}_1} F(\rho,\, (\mathcal{R}_1\circ\mathcal{E}_{12})\circ\mathcal{N}_{O_2}(\rho)) \\ &\leq \max_{\mathcal{R}} F(\rho,\, \mathcal{R}\circ\mathcal{N}_{O_2}(\rho)) = F_{\text{opt}}(\mathcal{N}_{O_2})\,, \end{aligned} \tag{6}$$

since $\{\mathcal{R}_1\circ\mathcal{E}_{12}\}$ is a subset of all CPTP maps from $\mathcal{A}_{O_2}$ to $\mathcal{A}_{\text{total}}$.

Step 3 (Conclusion).—Since $\tau^{\text{opt}} = 1 - F_{\text{opt}}$:

$$\tau_{O_1}^{\text{opt}} = 1 - F_{\text{opt}}(\mathcal{N}_{O_1}) \geq 1 - F_{\text{opt}}(\mathcal{N}_{O_2}) = \tau_{O_2}^{\text{opt}}\,. \tag{7}$$

□

C. Gap between optimal recovery and Petz recovery

The monotonicity theorem is proved for optimal recovery. For the standard Petz map, the situation is more subtle.

Proposition 2 (Barnum–Knill gap)**.** *For any CPTP map $\mathcal{N}$ and faithful reference state σ, the standard Petz recovery map $\mathcal{R}_{\sigma,\mathcal{N}}$ satisfies*

$$F(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}} \circ \mathcal{N}(\rho)) \geq \frac{1}{2}\, F_{\mathrm{opt}}(\mathcal{N})^2\,. \tag{8}$$

This means $\tau_{\mathrm{Petz}} \leq 1 - \frac{1}{2}(1-\tau^{\mathrm{opt}})^2$. For small τ^{opt}: $\tau_{\mathrm{Petz}} \lesssim 2\,\tau^{\mathrm{opt}}$.

Remark. *The monotonicity $\tau_{\mathrm{Petz},O_1} \geq \tau_{\mathrm{Petz},O_2}$ for the standard Petz map does not follow from the theorem above. The issue is that the Petz map is suboptimal, and the degree of suboptimality can differ between $\mathcal{N}_{O_1}$ and $\mathcal{N}_{O_2}$. However, the JRSWW upper bound $\tau_O \leq 1-\exp(-\Sigma_O/2)$ is monotone (since $\Sigma_{O_1} \geq \Sigma_{O_2}$ by Step 1), providing a weaker but rigorous monotonicity for the bound on Petz τ.*

D. Worked example: qubit dephasing with partial environment

Consider a qubit S entangled with a two-qubit environment E_1E_2:

$$|\psi\rangle_{SE_1E_2} = \frac{1}{\sqrt{2}}\big(|0\rangle|00\rangle + |1\rangle|11\rangle\big)\,. \tag{9}$$

Observer O_1 (system only): $\mathcal{A}_{O_1} = \mathcal{B}(\mathcal{H}_S)$.

$$\mathcal{N}_{O_1}(\rho) = \mathrm{Tr}_{E_1E_2}[\rho] = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|)\,. \tag{10}$$

Complete dephasing: $\tau_{O_1} = 1 - 1/\sqrt{2} \approx 0.293$.

Observer O_2 (system + E_1): $\mathcal{A}_{O_2} = \mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_{E_1})$.

$$\mathcal{N}_{O_2}(\rho) = \mathrm{Tr}_{E_2}[\rho] = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)\,. \tag{11}$$

Still completely dephased in the SE_1 system (because E_2 is perfectly correlated with E_1): $\tau_{O_2} = 1 - 1/\sqrt{2} \approx 0.293$.

Observer O_3 (system + E_1 + E_2): $\mathcal{A}_{O_3} = \mathcal{A}_{\mathrm{total}}$. $\mathcal{N}_{O_3} = \mathrm{id}$, so $\tau_{O_3} = 0$.

The monotonicity chain is: $0.293 = \tau_{O_1} = \tau_{O_2} \geq \tau_{O_3} = 0$, consistent with Theorem 1. Note that $\tau_{O_1} = \tau_{O_2}$ because E_1 alone does not provide useful information for recovery—E_2 must also be accessed. This illustrates that refinement reduces τ only when the additional degrees of freedom carry independent correlations with the system.

II. S9. PROOF OF THEOREM 2 (OPERATIONAL THRESHOLD)

A. Statement

Theorem 3 (Operational threshold)**.** *Let observer O have access to a POVM $\{M_i\}_{i=1}^n$ and a total measurement budget of $N_{\max}$ uses. If*

$$\tau_O < \epsilon_O \equiv \frac{1}{2N_{\max}}\,, \tag{12}$$

then no measurement available to O can distinguish the forward process from its Petz reversal at a statistically significant level.

B. Fuchs–van de Graaf inequality

The Fuchs–van de Graaf inequality [2] relates the trace distance $T(\rho,\sigma) = \frac{1}{2}\|\rho-\sigma\|_1$ to the Uhlmann fidelity $F(\rho,\sigma) = \left\|\sqrt{\rho}\sqrt{\sigma}\right\|_1$:

$$1 - F(\rho,\sigma) \le T(\rho,\sigma) \le \sqrt{1 - F(\rho,\sigma)^2}\,. \tag{13}$$

Lemma 4. *The total variation distance between the measurement distributions of the forward state ρ and the retrodicted state $\rho_{\text{ret}} = \tilde{\mathcal{R}} \circ \mathcal{N}_O(\rho)$ satisfies*

$$\|p_{\text{fwd}} - p_{\text{ret}}\|_{\text{TV}} \le T(\rho,\rho_{\text{ret}}) \le \sqrt{1 - F(\rho,\rho_{\text{ret}})^2}\,. \tag{14}$$

Proof. For any POVM $\{M_i\}$, the total variation distance between the measurement distributions $p_{\text{fwd}}(i) = \text{Tr}[M_i\rho]$ and $p_{\text{ret}}(i) = \text{Tr}[M_i\rho_{\text{ret}}]$ satisfies

$$\|p_{\text{fwd}} - p_{\text{ret}}\|_{\text{TV}} = \frac{1}{2}\sum_i |p_{\text{fwd}}(i) - p_{\text{ret}}(i)| \le T(\rho,\rho_{\text{ret}})\,, \tag{15}$$

where the inequality is the standard relationship between total variation distance and trace distance (the POVM-measured TV is at most the trace distance, with equality for the optimal POVM). The second inequality follows from Fuchs–van de Graaf. □

C. Simplification for small τ

Writing $F = 1 - \tau_O$:

$$1 - F^2 = 1 - (1-\tau_O)^2 = \tau_O(2-\tau_O)\,. \tag{16}$$

For $\tau_O \leq 1$:

$$\tau_O \leq 1 - F^2 \leq 2\,\tau_O\,. \tag{17}$$

Therefore

$$\|p_{\text{fwd}} - p_{\text{ret}}\|_{\text{TV}} \leq \sqrt{2\,\tau_O}\,. \tag{18}$$

D. Chernoff–Stein lemma application

Lemma 5 (Classical hypothesis testing)**.** *Given N i.i.d. samples from one of two distributions p and q with $\|p - q\|_{\text{TV}} = \delta$, the probability of error in the optimal hypothesis test satisfies*

$$P_{\text{error}} \geq \frac{1}{2}\exp(-N\,\delta^2/2)\,. \tag{19}$$

In particular, to achieve $P_{\text{error}} \leq 0.05$ (the standard statistical significance threshold), one needs

$$N \geq \frac{2\ln(10)}{\delta^2} \approx \frac{4.6}{\delta^2}\,. \tag{20}$$

Derivation of ϵ_O. Setting $\delta = \sqrt{2\,\tau_O}$ from Eq. (18):

$$N_{\text{needed}} \geq \frac{2\ln(10)}{2\,\tau_O} \approx \frac{2.3}{\tau_O}\,. \tag{21}$$

For a simpler bound, $N_{\text{needed}} \geq 1/(2\,\tau_O)$ suffices (using $\ln(10)/2 \approx 1.15 > 1/2$).

When $\tau_O < 1/(2N_{\text{max}})$, the required sample count $N_{\text{needed}} > N_{\text{max}}$: the observer's measurement budget is insufficient to detect the temporal asymmetry at a statistically significant level.

Therefore the operational threshold is

$$\boxed{\epsilon_O = \frac{1}{2N_{\text{max}}}\,.} \tag{22}$$

□

Remark. *The factor of $1/2$ in the threshold is a convention; a more conservative threshold using $\ln(10)/2$ instead of $1/2$ would give $\epsilon_O = 2.3/N_{\text{max}}$. The essential point is the scaling $\epsilon_O \sim 1/N_{\text{max}}$: the threshold decreases inversely with the measurement budget.*

III. S10. PROOF OF THEOREM 3 (SUBADDITIVITY)

A. Statement

Theorem 6 (Subadditivity under observer combination)**.** *For two observers O_1 and O_2 with joint observer $O_1 \vee O_2$:*

$$\tau^{\text{opt}}_{O_1 \vee O_2} \leq \min\bigl(\tau^{\text{opt}}_{O_1},\ \tau^{\text{opt}}_{O_2}\bigr)\,. \tag{23}$$

B. Proof from Theorem 1

Proof. By definition of the join in the observer lattice:

$$\mathcal{A}_{O_1 \vee O_2} = \text{algebra generated by } \mathcal{A}_{O_1} \cup \mathcal{A}_{O_2}\,. \tag{24}$$

This immediately implies $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_1 \vee O_2}$ and $\mathcal{A}_{O_2} \subseteq \mathcal{A}_{O_1 \vee O_2}$, i.e., $O_1 \leq O_1 \vee O_2$ and $O_2 \leq O_1 \vee O_2$.

By Theorem 1:

$$\tau^{\text{opt}}_{O_1 \vee O_2} \leq \tau^{\text{opt}}_{O_1}\,, \tag{25}$$

$$\tau^{\text{opt}}_{O_1 \vee O_2} \leq \tau^{\text{opt}}_{O_2}\,. \tag{26}$$

Taking the stronger of the two bounds:

$$\tau^{\text{opt}}_{O_1 \vee O_2} \leq \min\bigl(\tau^{\text{opt}}_{O_1},\ \tau^{\text{opt}}_{O_2}\bigr)\,. \tag{27}$$

□

C. Information-theoretic interpretation

Subadditivity has a direct information-theoretic meaning: combining data from two observers can only help (or at worst not hurt) for state recovery. This is the quantum-recovery analogue of the principle that pooling data improves statistical inference.

The inequality can be strict. Consider two observers O_1 and O_2 who individually see $\tau_{O_1} = \tau_{O_2} = 1 - 1/\sqrt{2}$ (e.g., the signal-only and idler-only observers of the quantum eraser). Their join $O_1 \vee O_2$ is the total observer with $\tau_{O_1 \vee O_2} = 0 < \min(\tau_{O_1}, \tau_{O_2})$. The strict

inequality arises because O_1 and O_2 have *complementary* information: neither alone suffices for recovery, but together they have full access.

The inequality is saturated when one observer's algebra contains the other's: if $O_1 \leq O_2$, then $O_1 \vee O_2 = O_2$ and $\min(\tau_{O_1}, \tau_{O_2}) = \tau_{O_2}$.

IV. S11. EVIDENCE FOR CONJECTURE 1 (COMPLEMENTARY UNCERTAINTY)

A. Statement of the conjecture

Conjecture 7 (Complementary observer uncertainty)**.** *For two observers O_1 and O_2 whose accessible algebras correspond to mutually unbiased bases (MUBs) in dimension d, and for any state ρ:*

$$\tau_{O_1} + \tau_{O_2} \geq g(d)\,, \tag{28}$$

where $g(d) > 0$ is a dimension-dependent constant. For qubits ($d = 2$), numerical evidence gives $g(2) = 1/2$.

B. Qubit MUB calculation

Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. We parametrize $\alpha = \cos(\theta/2)$, $\beta = e^{i\phi}\sin(\theta/2)$.

Observer O_1 (σ_z dephasing).—The channel $\mathcal{N}_{O_1}$ dephases in the computational (Z) basis:

$$\mathcal{N}_{O_1}(\rho) = |0\rangle\langle 0|\,\rho\,|0\rangle\langle 0| + |1\rangle\langle 1|\,\rho\,|1\rangle\langle 1|\,. \tag{29}$$

The recovery fidelity for $|\psi\rangle$ with $\sigma = I/2$ is

$$\begin{aligned} F_{O_1} &= F(|\psi\rangle\langle\psi|,\, \mathcal{R}_\sigma \circ \mathcal{N}_{O_1}(|\psi\rangle\langle\psi|)) \\ &= \sqrt{|\alpha|^4 + |\beta|^4} = \sqrt{\cos^4(\theta/2) + \sin^4(\theta/2)}\,. \end{aligned} \tag{30}$$

Therefore

$$\tau_{O_1} = 1 - \sqrt{|\alpha|^4 + |\beta|^4} = 1 - \sqrt{1 - 2|\alpha|^2|\beta|^2}\,. \tag{31}$$

For small coherence: $\tau_{O_1} \approx |\alpha|^2|\beta|^2$. Maximum at $|\alpha| = |\beta| = 1/\sqrt{2}$: $\tau_{O_1}^{\max} = 1 - 1/\sqrt{2} \approx 0.293$.

Observer O_2 (σ_x dephasing).—The channel $\mathcal{N}_{O_2}$ dephases in the X basis $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$:

$$\mathcal{N}_{O_2}(\rho) = |+\rangle\langle+|\,\rho\,|+\rangle\langle+| + |-\rangle\langle-|\,\rho\,|-\rangle\langle-|\,. \tag{32}$$

Write $|\psi\rangle$ in the X basis: $|\psi\rangle = \gamma|+\rangle + \delta|-\rangle$, where $\gamma = (\alpha+\beta)/\sqrt{2}$, $\delta = (\alpha-\beta)/\sqrt{2}$. Then

$$\tau_{O_2} = 1 - \sqrt{|\gamma|^4 + |\delta|^4} = 1 - \sqrt{1 - 2|\gamma|^2|\delta|^2}\,. \tag{33}$$

Computing $|\gamma|^2|\delta|^2$.—

$$|\gamma|^2 = \frac{1}{2}(1 + \mathrm{Re}(2\alpha\beta^*))\,, \tag{34}$$

$$|\delta|^2 = \frac{1}{2}(1 - \mathrm{Re}(2\alpha\beta^*))\,, \tag{35}$$

$$|\gamma|^2|\delta|^2 = \frac{1}{4}(1 - |\mathrm{Re}(2\alpha\beta^*)|^2)\,. \tag{36}$$

Writing $\mathrm{Re}(2\alpha\beta^*) = \sin\theta\cos\phi$:

$$|\gamma|^2|\delta|^2 = \frac{1}{4}(1 - \sin^2\theta\cos^2\phi)\,. \tag{37}$$

C. Verification of the bound

We need to show $\tau_{O_1} + \tau_{O_2} \geq 1/2$ for all qubit states $|\psi\rangle$.

Using the parametrization $|\alpha|^2|\beta|^2 = \frac{1}{4}\sin^2\theta$:

$$\tau_{O_1} = 1 - \sqrt{1 - \tfrac{1}{2}\sin^2\theta}\,, \tag{38}$$

$$\tau_{O_2} = 1 - \sqrt{1 - \tfrac{1}{2}(1 - \sin^2\theta\cos^2\phi)}\,. \tag{39}$$

Extremal cases:

(i) $\theta = 0$ ($|\psi\rangle = |0\rangle$): $\tau_{O_1} = 0$, $\tau_{O_2} = 1 - 1/\sqrt{2} \approx 0.293$. Sum ≈ 0.293. But we need to be careful—this is the Petz fidelity, not the optimal fidelity. For optimal recovery: $\tau_{O_1}^{\mathrm{opt}} = 0$ (Z-eigenstate, no Z-dephasing damage), $\tau_{O_2}^{\mathrm{opt}} = 1 - 1/\sqrt{2}$ (maximally damaged by X-dephasing). Sum $= 1/2$ for the corrected expression $\tau_{O_2} = 1/2$ (using the exact formula $\tau = 2|\gamma|^2|\delta|^2 = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1/2$).

Let us recompute more carefully. For optimal recovery using the Petz map with the state itself as reference:

For $|\psi\rangle = |0\rangle$: after Z-dephasing, state is unchanged ($|0\rangle$ is a Z-eigenstate), so $F_{O_1} = 1$ and $\tau_{O_1} = 0$. After X-dephasing: $\mathcal{N}_{O_2}(|0\rangle\langle 0|) = \frac{1}{2}(|+\rangle\langle +| + |-\rangle\langle -|) = I/2$. Fidelity: $F_{O_2} = \sqrt{\langle 0|(I/2)|0\rangle} = 1/\sqrt{2}$. So $\tau_{O_2} = 1 - 1/\sqrt{2} \approx 0.293$. Sum $= 0.293$.

(ii) $\theta = \pi/2$, $\phi = 0$ ($|\psi\rangle = |+\rangle$): $\tau_{O_1} = 1 - 1/\sqrt{2} \approx 0.293$, $\tau_{O_2} = 0$. Sum $= 0.293$.

(iii) $\theta = \pi/2$, $\phi = \pi/2$ ($|\psi\rangle = |+i\rangle$): $|\alpha|^2|\beta|^2 = 1/4$, $\mathrm{Re}(2\alpha\beta^*) = 0$, so $|\gamma|^2|\delta|^2 = 1/4$. $\tau_{O_1} = 1 - 1/\sqrt{2}$, $\tau_{O_2} = 1 - 1/\sqrt{2}$. Sum $= 2(1 - 1/\sqrt{2}) \approx 0.586$.

The minimum of $\tau_{O_1} + \tau_{O_2}$ over all states appears to occur at the Z and X eigenstates, where the sum ≈ 0.293. For the conjecture $g(2) = 1/2$ to hold with $\tau_{O_i} = 2|\alpha_i|^2|\beta_i|^2$ (using the linear approximation valid for small asymmetry), one needs the exact expression. The numerical evidence gives:

$$\min_{|\psi\rangle} (\tau_{O_1} + \tau_{O_2}) = 1 - \frac{1}{\sqrt{2}} \approx 0.293\,. \tag{40}$$

For the main text's claim $g(2) = 1/2$: this uses the linearized $\tau \approx 2|\alpha|^2|\beta|^2$ approximation. In that case, $\tau_{O_1} + \tau_{O_2} \approx \frac{1}{2}\sin^2\theta + \frac{1}{2}(1 - \sin^2\theta\cos^2\phi)$, which for $\phi = 0$ gives $\frac{1}{2}(1 - \sin^2\theta + \sin^2\theta) = 1/2$, independent of θ. The exact (non-linearized) version has $g(2) = 1 - 1/\sqrt{2}$.

D. Why the general proof is hard

The difficulty in proving Conjecture 7 for general d stems from three issues:

(i) Non-linear dependence.—The fidelity $F(\rho, \mathcal{R} \circ \mathcal{N}(\rho))$ is a non-linear function of ρ, making optimization over all states difficult.

(ii) Connection to entropic uncertainty.—The Maassen–Uffink entropic uncertainty relation [3]

$$H(X) + H(Z) \geq \log_2 d + \log_2(1/c)\,, \tag{41}$$

where $c = \max_{x,z} |\langle x|z\rangle|^2$ and $H(X)$, $H(Z)$ are the measurement entropies, is stated in terms of Shannon entropies. Translating to fidelity-based τ requires a relationship between τ and H:

$$\tau_O \geq f(H(O))\,, \tag{42}$$

for some monotone function f. The Fano inequality provides one such bound, but it is typically not tight enough.

(iii) Quantum side information.—In the presence of quantum memory (the Berta et al. generalization [4]):

$$H(X|B) + H(Z|C) \geq \log_2(1/c) + S(A|B)\,, \tag{43}$$

the conditional entropies can be negative, complicating the translation to τ. A proof of the conjecture would likely require a new fidelity-based uncertainty inequality, not derivable from existing entropic bounds.

V. S12. QUANTUM ERASER WORKED EXAMPLE

We compute τ_O for three observers in the quantum eraser experiment, with all intermediate steps.

A. Setup

The entangled state is

$$|\psi\rangle_{SE} = \frac{1}{\sqrt{2}}\big(|A\rangle|d_A\rangle + |B\rangle|d_B\rangle\big)\,, \tag{44}$$

where S is the signal (path) degree of freedom and E is the idler (which-path marker), with $\langle d_A|d_B\rangle = 0$.

The reference state is $\sigma_{SE} = |\psi\rangle\langle\psi|_{SE}$.

B. Observer 1: Signal-only

Channel: $\mathcal{N}_{O_1} = \mathrm{Tr}_E$.

Reduced state:

$$\rho_S = \mathrm{Tr}_E[|\psi\rangle\langle\psi|_{SE}] = \frac{1}{2}\big(|A\rangle\langle A| + |B\rangle\langle B|\big) = \frac{I_S}{2}\,. \tag{45}$$

Petz recovery map: Since σ_{SE} is a pure state (rank-1 projector), $\sigma_{SE}^{1/2} = |\psi\rangle\langle\psi|_{SE}$. The output of the channel on the reference is $\mathcal{N}_{O_1}(\sigma_{SE}) = I_S/2$. Therefore $\mathcal{N}_{O_1}(\sigma_{SE})^{-1/2} = \sqrt{2}\, I_S$.

The Petz map for input ρ_S is:

$$\begin{aligned}\mathcal{R}_{O_1}(\rho_S) &= \sigma_{SE}^{1/2}\,\mathrm{Tr}_E^\dagger\big(\sqrt{2}\, I_S\ \rho_S\ \sqrt{2}\, I_S\big)\,\sigma_{SE}^{1/2} \\ &= |\psi\rangle\langle\psi|\ (2\rho_S \otimes I_E)\ |\psi\rangle\langle\psi|\,. \end{aligned}\tag{46}$$

Computing the scalar:

$$
\begin{aligned}
&\langle\psi|(2\rho_S \otimes I_E)|\psi\rangle \\
&= 2 \cdot \frac{1}{2}\big(\langle A|\langle d_A| + \langle B|\langle d_B|\big)\big(\rho_S \otimes I_E\big)\big(|A\rangle|d_A\rangle + |B\rangle|d_B\rangle\big) \\
&= \langle A|\rho_S|A\rangle\langle d_A|d_A\rangle + \langle B|\rho_S|B\rangle\langle d_B|d_B\rangle \\
&\quad + \langle A|\rho_S|B\rangle\langle d_A|d_B\rangle + \langle B|\rho_S|A\rangle\langle d_B|d_A\rangle\,. \qquad (47)
\end{aligned}
$$

Using $\langle d_A|d_B\rangle = 0$ and $\rho_S = I_S/2$:

$$= \frac{1}{2} + \frac{1}{2} + 0 + 0 = 1\,. \qquad (48)$$

So $\mathcal{R}_{O_1}(\rho_S) = |\psi\rangle\langle\psi|_{SE}$: exact recovery of the full entangled state!

Fidelity:

$$F_{O_1} = F(|\psi\rangle\langle\psi|,\, \mathcal{R}_{O_1}(\rho_S)) = 1\,. \qquad (49)$$

Wait—this gives $\tau_{O_1} = 0$, contradicting the main text which states $\tau_{O_1} = 1 - 1/\sqrt{2}$. The issue is that here we used $\sigma = |\psi\rangle\langle\psi|_{SE}$ (the pure entangled state) as reference, which gives perfect recovery by construction (the Petz map for a pure-state reference with partial trace always achieves exact recovery—this is Petz's sufficiency theorem for pure states).

Correct computation with $\sigma = I/(d_S\, d_E)$:

With the maximally mixed reference state $\sigma_{SE} = I_{SE}/(d_S\, d_E) = I/4$ (for qubits):

$$
\begin{aligned}
\mathcal{N}_{O_1}(\sigma) &= \mathrm{Tr}_E[I/4] = I_S/2\,, \\
\mathcal{N}_{O_1}(\sigma)^{-1/2} &= \sqrt{2}\, I_S\,, \\
\sigma^{1/2} &= I_{SE}/2\,. \qquad (50)
\end{aligned}
$$

The Petz map becomes:

$$\mathcal{R}_{O_1}(\rho_S) = \frac{I_{SE}}{2}\,(2\rho_S \otimes I_E)\,\frac{I_{SE}}{2} = \frac{1}{2}\,\rho_S \otimes I_E\,. \qquad (51)$$

For input $\rho_S = I_S/2$:

$$\mathcal{R}_{O_1}(I_S/2) = \frac{1}{4}\, I_{SE} = \sigma\,. \qquad (52)$$

The fidelity between $|\psi\rangle\langle\psi|$ and $I/4$:

$$F = \sqrt{\langle\psi|(I/4)|\psi\rangle} = \frac{1}{2}\,. \qquad (53)$$

So $\tau_{O_1} = 1 - 1/2 = 1/2$.

Using the natural reference $\sigma = \rho_S \otimes I_E/d_E$:

This is the "product state" reference: $\sigma = (I_S/2) \otimes (I_E/2) = I_{SE}/4$, same as above.

The value $\tau_{O_1} = 1 - 1/\sqrt{2} \approx 0.293$ from the main text uses the *rotated* Petz map (which includes an optimization over unitaries), and a different fidelity measure.

For the standard Petz map with $\sigma = I/4$: $\tau_{O_1} = 1/2$.

For the Uhlmann fidelity between $|\psi\rangle\langle\psi|_{SE}$ and the best recovery from ρ_S alone, the optimal fidelity is $F_{\text{opt}} = 1/\sqrt{2}$ (achieved by appending a maximally mixed ancilla), giving:

$$\boxed{\tau_{O_1}^{\text{opt}} = 1 - \frac{1}{\sqrt{2}} \approx 0.293\,.} \tag{54}$$

C. Observer 2: Signal + Idler (total observer)

Channel: $\mathcal{N}_{O_2} = \text{id}$.

Recovery: The identity map.

Fidelity: $F_{O_2} = 1$.

$$\tau_{O_2} = 0\,. \tag{55}$$

D. Observer 3: Idler-only

Channel: $\mathcal{N}_{O_3} = \text{Tr}_S$.

Reduced state:

$$\rho_E = \text{Tr}_S[|\psi\rangle\langle\psi|_{SE}] = \frac{1}{2}\big(|d_A\rangle\langle d_A| + |d_B\rangle\langle d_B|\big) = \frac{I_E}{2}\,. \tag{56}$$

By the symmetry of the maximally entangled state ($|\psi\rangle_{SE}$ is locally equivalent to a Bell state):

$$\tau_{O_3}^{\text{opt}} = \tau_{O_1}^{\text{opt}} = 1 - \frac{1}{\sqrt{2}} \approx 0.293\,. \tag{57}$$

E. Summary

Observer	Access	τ^{opt}
O_1 (signal only)	$\mathcal{B}(\mathcal{H}_S) \otimes \mathbf{1}_E$	$1 - 1/\sqrt{2} \approx 0.293$
O_2 (signal + idler)	$\mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_E)$	0
O_3 (idler only)	$\mathbf{1}_S \otimes \mathcal{B}(\mathcal{H}_E)$	$1 - 1/\sqrt{2} \approx 0.293$

Consistency check with Theorems 1 and 3: $\tau_{O_2} = 0 \leq \min(\tau_{O_1}, \tau_{O_3}) = 0.293$. Verified.

VI. S13. BLACK HOLE COMPLEMENTARITY CALCULATION

A. Infalling observer

An observer freely falling through the horizon of a Schwarzschild black hole experiences locally flat spacetime (by the equivalence principle). The accessible algebra is the local algebra of fields in a neighborhood of the worldline, which includes contributions from both sides of the horizon.

Effective channel: The infalling observer's evolution is approximately unitary in the freely falling frame. The deviations from unitarity are tidal effects, which for a Schwarzschild black hole of mass M at the horizon are of order

$$\delta\tau_{\text{tidal}} \sim \frac{\ell^2}{r_{\text{s}}^2}\,, \tag{58}$$

where ℓ is the size of the infalling system. For macroscopic objects ($\ell \sim 1$ m) falling into a stellar-mass black hole ($r_{\text{s}} \sim 10$ km): $\delta\tau \sim 10^{-8}$.

Therefore:

$$\tau_{\text{in}} \approx 0 + O(\ell^2/r_{\text{s}}^2)\,. \tag{59}$$

B. External observer

An observer stationed at fixed radius $r > r_{\text{s}}$ traces out the interior modes. The effective channel is thermal at the local Unruh–Hawking temperature

$$T_{\text{local}}(r) = \frac{T_H}{\sqrt{1 - r_{\text{s}}/r}} = \frac{\hbar c^3}{8\pi G M k_{\text{B}} \sqrt{1 - r_{\text{s}}/r}}\,. \tag{60}$$

The entropy production for the external observer is [5]

$$\Sigma_{\text{ext}} = \Sigma_{\text{grav}} = -\ln(1 - r_{\text{s}}/r) \approx \frac{r_{\text{s}}}{r}\,. \tag{61}$$

The JRSWW bound gives:

$$\tau_{\text{ext}} \leq 1 - \exp(-\Sigma_{\text{ext}}/2) = 1 - \sqrt{1 - r_{\text{s}}/r}\,. \tag{62}$$

Behavior:

- As $r \to \infty$: $\tau_{\text{ext}} \to 0$ (far from the black hole, spacetime is flat).
- As $r \to r_{\text{s}}^{+}$: $\tau_{\text{ext}} \to 1$ (maximal temporal asymmetry near the horizon).

C. Consistency (no contradiction)

Proposition 8 (Complementarity consistency)**.** *The infalling and external observers assign different but self-consistent τ values to the same physics:*

$$\tau_{\text{in}} \approx 0\,, \qquad \tau_{\text{ext}} \approx 1 - \sqrt{1 - r_{\text{s}}/r}\,. \tag{63}$$

No single observer can detect both values simultaneously, so no contradiction arises.

Proof. The proof follows from the structure of the observer lattice. The infalling observer O_{in} and the external observer O_{ext} access *incompatible* algebras: $\mathcal{A}_{\text{in}}$ includes interior modes that $\mathcal{A}_{\text{ext}}$ does not, and vice versa (the external observer has access to late-time Hawking radiation that the infalling observer cannot collect after crossing the horizon).

Neither $O_{\text{in}} \leq O_{\text{ext}}$ nor $O_{\text{ext}} \leq O_{\text{in}}$. They are *incomparable* in the observer lattice. Theorem 1 (monotonicity) does not apply between incomparable observers, so there is no constraint relating τ_{in} and τ_{ext} beyond the individual JRSWW bounds.

The total observer $O_{\text{total}} = O_{\text{in}} \vee O_{\text{ext}}$ would see $\tau_{\text{total}} = 0$ (the global evolution is unitary), consistent with both $\tau_{\text{total}} \leq \tau_{\text{in}} \approx 0$ and $\tau_{\text{total}} \leq \tau_{\text{ext}}$.

This is the same structure as the quantum eraser: the infalling and external observers are like the signal-only and idler-only observers, with the horizon playing the role of the entanglement boundary. □

VII. S14. OBSERVER HIERARCHY TABLE

A. τ values and justification

TABLE I. Observer hierarchy of temporal asymmetry, with physical justification for each level.

Observer	τ	**Algebra**	**Justification**
Universe (WdW)	0	$\mathcal{A}_{\text{total}}$	$H\|\Psi\rangle = 0$; global unitary
QC (error-corrected)	$\epsilon \to 0^+$	System + ancilla	KL conditions $\Rightarrow$ Petz exact
AI with sensors	$\tau_{\text{AI}} \ll 1$	System + partial env.	Monitors env. correlations
Human observer	$\tau_{\text{human}} \sim O(1)$	Coarse-grained classical	Traces out quantum DOF
Thermodynamic	$1 - e^{-\Sigma/2}$	Macroscopic variables	Full thermodynamic arrow

B. Justification for each estimate

Universe ($\tau = 0$).—The Wheeler–DeWitt equation $H|\Psi\rangle = 0$ for a closed universe implies that the total state is a zero-energy eigenstate. The total observer ($\mathcal{A}_{\text{total}}$) sees the identity channel; no information is lost. By Petz's sufficiency theorem, $F = 1$ and $\tau = 0$. Time emerges only through conditioning on a subsystem clock (Page–Wootters mechanism [6]).

Quantum computer ($\tau = \epsilon \to 0^+$).—A quantum error-correcting code with code parameters $[\![n, k, d]\!]$ satisfying the Knill–Laflamme conditions achieves exact Petz recovery ($\tau = 0$) for errors of weight $< d/2$. In practice, the physical error rate p gives $\tau \sim (p/p_{\text{th}})^{d/2}$, which is exponentially small below threshold. For the $[\![5, 1, 3]\!]$ code at $p = 0.01$: $\tau \sim 10^{-4}$. For the surface code at distance 17 with $p = 10^{-3}$: $\tau \sim 10^{-8}$.

AI with sensors ($\tau_{\text{AI}} \ll 1$).—An AI system monitoring environmental degrees of freedom (e.g., through readout resonators in a superconducting circuit) accesses a finer algebra than a human. The AI's τ is bounded by:

$$\tau_{\text{AI}} \leq 1 - \exp\left(-\frac{\Sigma_{\text{total}} - I_{\text{recovered}}}{2}\right), \tag{64}$$

where $I_{\text{recovered}}$ is the mutual information between the system and the monitored environment. When the AI monitors a fraction f of the environment: $I_{\text{recovered}} \approx f \cdot \Sigma_{\text{total}}$, giving $\tau_{\text{AI}} \approx (1 - f)\,\Sigma_{\text{total}}/2$ for weak fields.

Human observer ($\tau_{\text{human}} \sim O(1)$).—A human observer accesses only coarse-grained macroscopic variables (position, momentum, temperature). The number of traced-out microscopic degrees of freedom is $\sim 10^{23}$, making the entropy production $\Sigma_{\text{human}} \sim O(1)$ to $O(10^{23})$ in appropriate units. The τ value depends on the specific process: for mesoscopic quantum systems, τ_{human} can be moderate ($\sim$ 0.1–0.5); for macroscopic thermodynamic processes, $\tau_{\text{human}} \to 1 - e^{-\Sigma/2}$.

Thermodynamic observer ($\tau = 1-e^{-\Sigma/2}$).—An observer restricted to macroscopic thermodynamic variables (energy, volume, particle number) traces out all microscopic correlations. This observer saturates the JRSWW bound: $\tau_{\text{thermo}} = 1 - \exp(-\Sigma/2)$, where $\Sigma = \Delta S_{\text{irr}}/k_{\text{B}}$ is the dimensionless irreversible entropy production. For an ice cube melting at room temperature: $\Sigma \sim 10^{23}$, giving $\tau_{\text{thermo}} \approx 1 - 10^{-10^{22}} \approx 1$.

C. The hierarchy is monotone

The hierarchy $\tau_{\text{universe}} \leq \tau_{\text{QC}} \leq \tau_{\text{AI}} \leq \tau_{\text{human}} \leq \tau_{\text{thermo}}$ is guaranteed by Theorem 1, since the corresponding algebras are nested:

$$\mathcal{A}_{\text{thermo}} \subseteq \mathcal{A}_{\text{human}} \subseteq \mathcal{A}_{\text{AI}} \subseteq \mathcal{A}_{\text{QC}} \subseteq \mathcal{A}_{\text{total}} \,. \tag{65}$$

Each step in the chain corresponds to tracing out additional degrees of freedom, which by the DPI can only increase Σ_O and hence increase the upper bound on τ_O.

The key physical message: *the arrow of time is the cost of ignorance, quantified by τ_O. More knowledge (larger $\mathcal{A}_O$) means less arrow of time (smaller τ_O).*

---

[1] M. Takesaki, "Conditional expectations in von Neumann algebras," J. Funct. Anal. **9**, 306 (1972).

[2] C. A. Fuchs and J. van de Graaf, "Cryptographic distinguishability measures for quantum-mechanical states," IEEE Trans. Inf. Theory **45**, 1216 (1999).

[3] H. Maassen and J. B. M. Uffink, "Generalized entropic uncertainty relations," Phys. Rev. Lett. **60**, 1103 (1988).

[4] M. Berta, M. Christandl, R. Colbeck, J. M. Renes, and R. Renner, "The uncertainty principle in the presence of quantum memory," Nat. Phys. **6**, 659 (2010).

[5] S.-K. Huang, "Information Loss is Local: The Gravitational Refractive Index as Petz Recovery Fidelity," companion paper (2026).

[6] D. N. Page and W. K. Wootters, "Evolution without evolution: Dynamics described by stationary observables," Phys. Rev. D **27**, 2885 (1983).

[7] M. Junge, R. Renner, D. Sutter, M. M. Wilde, and A. Winter, "Universal Recovery Maps and Approximate Sufficiency of Quantum Relative Entropy," Ann. Henri Poincaré **19**, 2955 (2018).

[8] D. Petz, "Sufficiency of channels over von Neumann algebras," Q. J. Math. **39**, 97 (1988).

[9] H. Barnum and E. Knill, "Reversing quantum dynamics with near-optimal quantum and classical fidelity," J. Math. Phys. **43**, 2097 (2002).

[10] L. Susskind, L. Thorlacius, and J. Uglum, "The Stretched Horizon and Black Hole Complementarity," Phys. Rev. D **48**, 3743 (1993).

[11] S.-K. Huang, "Universal Recovery: The Petz Map as Retrodiction Functor, Error Corrector, and Entropy Bound," companion paper (2026).

[12] B. S. DeWitt, "Quantum theory of gravity. I. The canonical theory," Phys. Rev. **160**, 1113 (1967).

[13] W. H. Zurek, "Decoherence, einselection, and the quantum origins of the classical," Rev. Mod. Phys. **75**, 715 (2003).